

# Where the Profit Leaks

**\$2.30M**

Sales, 2014 to 2017

**\$286K**

Profit, 2014 to 2017

**12.5%**

Profit margin

**+89%**

Annual profit growth, 2014 to 2017

## THE STORY IN BRIEF

On the surface, this is a success story: profit almost doubled in four years. Underneath, the same data describes a business that gives away part of what it earns. Discounts beyond 20% turn profitable orders into losses, two regions shrank in the very year the company grew and a group of customers who look loyal costs more than it brings in. None of these leaks threatens the business today. Together, they explain why a company with \$2.3M in sales keeps only 12.5 cents of every dollar.

## OVERVIEW

# The story in seven steps

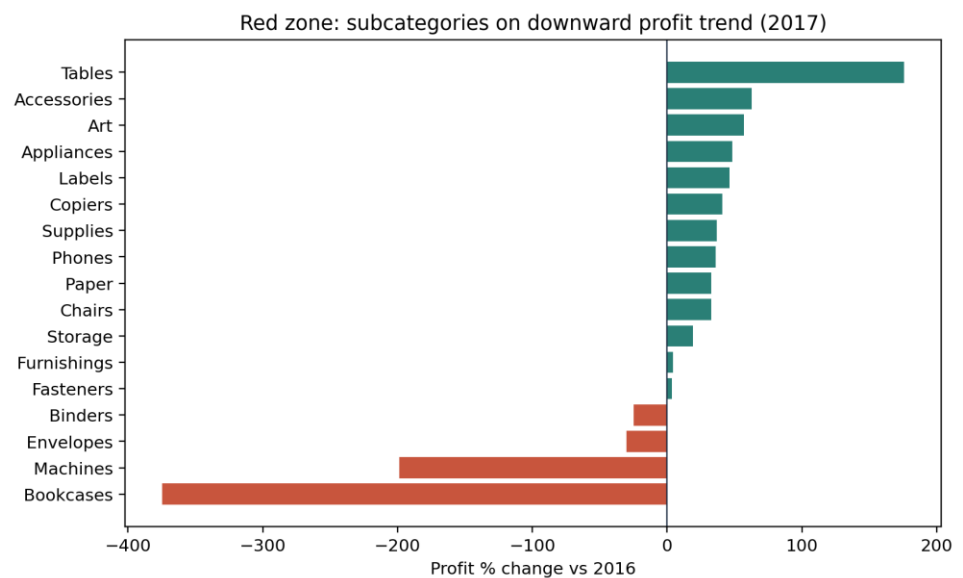
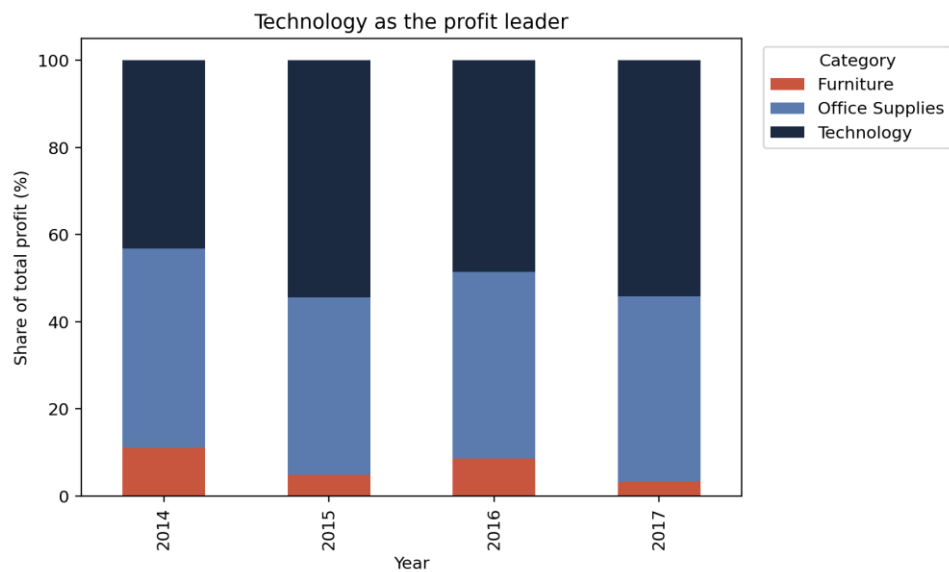
The analysis starts with what is happening, moves to where and why it happens, then to who drives it, and ends with how the business should monitor it.

| CHAPTER                        | WHAT THE DATA SHOWS                                                          | WHY IT MATTERS                                          |
|--------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>1. Growth</b>               | Profit rose 89% to \$93.3K, carried by Technology and Office Supplies.       | Furniture is volatile and fell to 3% of profit in 2017. |
| <b>2. The 2017 paradox</b>     | Company profit grew 14.2% while Central fell 63% and South 50%.              | Consolidated growth hid a regional contraction.         |
| <b>3. Central</b>              | A 7.9% margin and 47.6% of regional profit from three customers.             | The region carries both pricing risk and customer risk. |
| <b>4. The 20% line</b>         | Average margin drops from +26.7% to -91.5% once discounts pass 20 to 30%.    | Discounting has a break point not a gradual cost.       |
| <b>5. Seventeen thresholds</b> | The break point differs by subcategory.                                      | A single discount cap is the wrong instrument.          |
| <b>6. Customers</b>            | 19% of customers make 80% of profit; 137 active customers lose \$189 each.   | Volume metrics hide who really creates value.           |
| <b>7. Warning lights</b>       | Extreme margins mean noise on tiny orders and real losses on mid sized ones. | Alerts only work when read alongside order size.        |

## A business that grows with different patterns

Between 2014 and 2017, annual profit climbed from \$49.5K to \$93.3K, an increase of 89%. Technology drove most of it, while office supplies grew steadily alongside it.

However, furniture is the least stable of the three categories, falling to 45% in 2015, rising 131% in 2016 and falling 57% again in 2017, when it contributed only \$3.0K, about 3% of total profit. Most of that last drop came from a single subcategory, bookcases. The lack of contribution is not due to discount movement as its average remains at 17%.

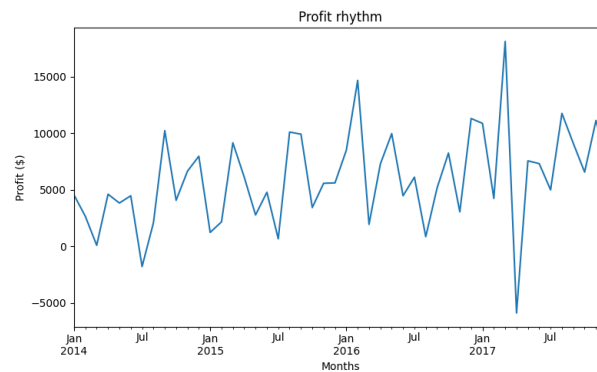


## The 2017 paradox

2017 as a record year: Company profit rose 14.2%. That headline, however, hides two regions moving in the opposite direction. Central fell by 63% and in South by 50%, while East grew by 65% and West by 82%. The weakness was not a general slowdown.

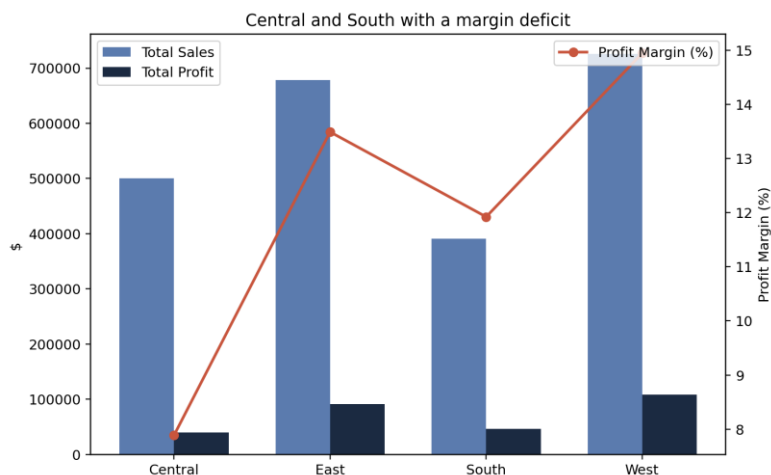
| Region     | Central | East  | South  | West  |
|------------|---------|-------|--------|-------|
| Order Year |         |       |        |       |
| 2015       | 2431.1% | 23.7% | -30.0% | 2.1%  |
| 2016       | 69.8%   | -4.5% | 111.2% | 17.4% |
| 2017       | -63.0%  | 65.0% | -49.6% | 82.1% |

Aggregate growth is a poor early warning system. A manager reading only the company total at the start of 2018 would have seen the best year on record and missed that half of the regional network was losing ground.



## Central, a region running on thin margins

Central is the only region where furniture ends the four years in loss, close to \$3K in total. It also converts sales into profit less efficiently than any other region.



The second weakness is less visible and potentially more serious. Three customers, out of the 629 who bought in central region, generated 47.6% of the region's profit. The 2017 fall in Central was driven by bookcases rather than by the loss of any of these customers, but the exposure remains: losing the largest of the three alone would remove at least 16% of the region's profit.

West shows what a healthy region looks like: the highest sales, the highest profit, the lowest average discount and the lowest dependence on individual customers.

## CHAPTER 4

# The 20% line

**+26.7%**

Average margin on orders discounted up to 20%

**-91.5%**

Average margin on orders discounted above 30%

The relationship between discount and profit is not a gentle slope. Up to 20%, most orders remain profitable. Beyond 30%, almost all of them lose money, and the losses grow with the discount until they reach the largest single loss in the dataset: \$6,600 on one order line sold at 70% off.

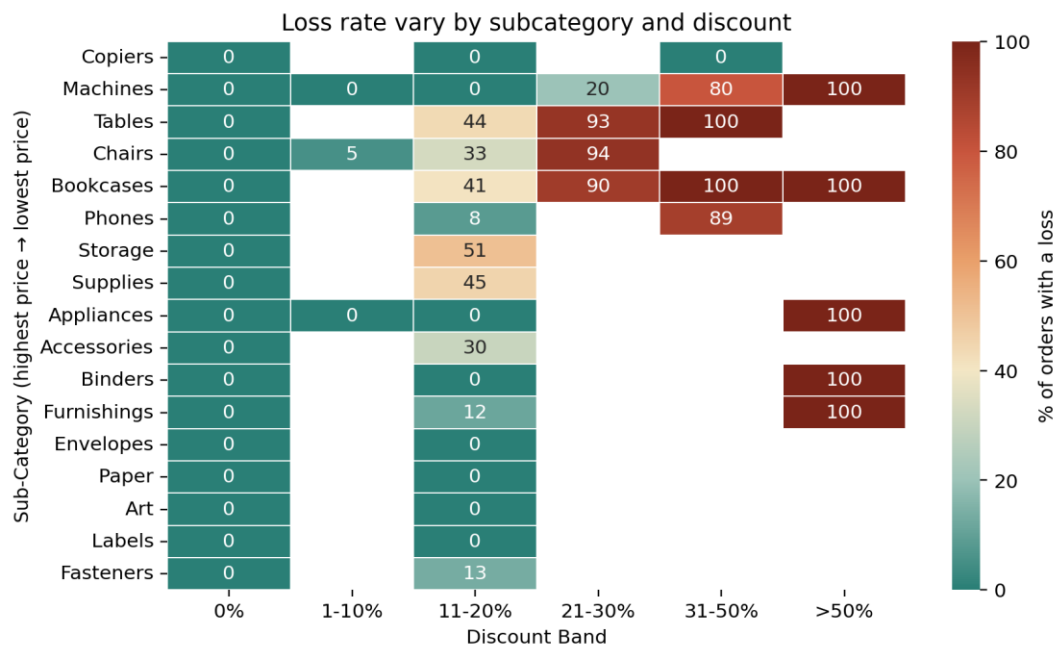
This shape matters for how the business is managed. A correlation coefficient would average both regimes into a single slope and suggest that each extra point of discount costs a little profit. The reality is a break point. Below it, discounts are an affordable sales tool. Above it, they destroy value.



The line does not sit in the same place for every category. Furniture margins turn negative in the 21 to 30% band. Technology remains profitable through that same band and only falls into loss above 30%. Office supplies follows its own pattern.

## One company, seventeen thresholds

Breaking the analysis down by subcategory confirms that the safe level of discount belongs to each product line rather than to the company as a whole. Tables, chairs and bookcases, among the most expensive items in the catalogue, sell at a loss on 90% or more of their order lines once the discount reaches 21 to 30%. Phones and machines hold up longer, but between 31% and 50% off, 89% and 80% of their lines respectively lose money.

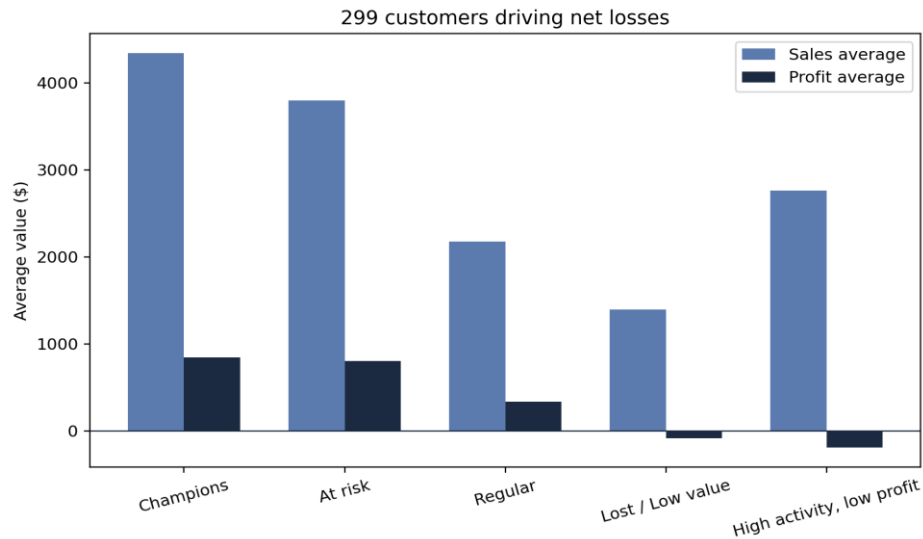


And every subcategory that is ever discounted above 50% loses money on every single line sold at that level.

## Who the profit belongs to

Profit is highly concentrated: 19% of customers generate 80% of it. This classic Pareto pattern makes retaining the top tier a financial priority rather than a marketing preference.

To look beyond volume, customers were segmented by recency, frequency and a monetary score based on profit. Three segments build the company's profit. Champions contribute \$128.1K and regular customers \$55.1K. The largest block, \$142.9K, comes from an at risk group of 178 customers who have been highly profitable but have not bought recently.



Two segments take profit away. Lost or low value customers cost \$14.2K, which is to be expected. The surprise is a group of 137 customers with high activity and low profit. They order almost as often as Champions, 8.4 orders on average against 8.5, and each spends around \$2,759, yet they lose \$189 per customer, \$25.9K in total.

## CHAPTER 7

# Reading the warning lights correctly

At order level, the worst margins in the dataset reach -275%. Taken at face value, they would all look like emergencies. They are not the same kind of problem.

| ORDER          | SALES    | PROFIT    | MARGIN | WHAT IT MEANS                            |
|----------------|----------|-----------|--------|------------------------------------------|
| US-2017-155299 | \$1.62   | -\$4.47   | -275%  | Ratio effect on a tiny order             |
| CA-2017-165099 | \$1.39   | -\$3.76   | -270%  | Ratio effect on a tiny order             |
| CA-2015-164007 | \$145.50 | -\$392.86 | -270%  | <b>Real loss driven by deep discount</b> |
| CA-2014-130421 | \$176.77 | -\$459.61 | -260%  | <b>Real loss driven by deep discount</b> |

Orders worth a few dollars reach extreme percentages because a loss of a few dollars is large relative to their size. Mid sized orders reach the same percentages for a very different reason: heavy discounting that costs the business hundreds of dollars per order. The same number signals noise in one case and a recoverable loss in the other, so any margin alert the business sets up should be split by order size and point attention to the orders where money is actually at stake.

## Five decisions for the year ahead

The decisions below follow the order of the story.

1. Replace the flat discount ceiling with limits by subcategory. Cap furniture at 20% and technology at 30%, remove discounts above 50% and review storage and supplies at 11 to 20%.
2. Win back at risk segment. 178 customers hold the largest block of profit but have not bought recently. Launch a retention campaign ranked by each's customer historical profit.
3. Reprice the high activity, low profit accounts. 137 customers lose 25.9k\$ in total. Review discount terms and product mix account by account.
4. Reduce central region's exposure. Build account plans for the top three customers and review furniture pricing and costs in the region.
5. Split margin alerts by order size. Extreme margins are ratio effects on tiny orders. Flag only orders above a minimum sales value for review

### LIMITATIONS

The findings are robust within the data available, but four boundaries should shape how they are used. First, profit arrives as a single figure with no breakdown of product cost or freight, and returns are not recorded at all, so the analysis shows where money is lost more reliably than why, and profit may be overstated wherever returns are frequent. Second, the data is observational: discounts may be applied to products that were already hard to sell, so the proposed caps should be validated with a controlled test before a full rollout. Third, customer segments depend on reasonable but adjustable thresholds and on names as customer identifiers, so their sizes are estimates. Fourth, some results rest on few observations, notably Central's reliance on three customers.

### ABOUT THE ANALYSIS

The analysis was carried out in Python with pandas, NumPy, SciPy, matplotlib and seaborn. The raw file contained 9,994 order lines. Two exact duplicates and five lines with missing financial values, all belonging to a single product with no other records, were removed, leaving 9,987 lines. Outliers were kept on purpose, since large orders and deep discounts are real transactions and central to the findings. The full notebook, with every step and decision documented, is available in the project repository.